

ECM3412

UNIVERSITY OF EXETER

**COLLEGE OF ENGINEERING,
MATHEMATICS AND
PHYSICAL SCIENCES**

COMPUTER SCIENCE

MAY/JUNE 2012

Nature Inspired Computation

Module Leader: Dr Edward Keedwell

Duration: 2 HOURS

Answer Question 1, and any two of the **THREE** other questions.

Question 1, is worth **40 marks**.

Other Questions are worth **30 marks** each.

Candidates are advised to spend **FORTY-FIVE** minutes on Question 1,

And **THIRTY-FIVE** minutes on other questions.

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examinations.

This is a **CLOSED BOOK** examination.

Compulsory question

1. (a) Write pseudocode for *ant colony optimisation* (ACO). In your answer, explain how the main operations of the algorithm allow it to find good solutions within reasonable time.

(8 marks)

- (b) Explain what is meant by an *approximate algorithm* and why this type of algorithm is often required in large real-world problems.

(6 marks)

- (c) Most population-based evolutionary algorithms require a selection method to determine which individuals are selected for variation and propagation to the next generation.

(i) Briefly explain why selectors should not always select the very fittest individuals in the population.

(ii) Show (using a diagram if necessary) how a *roulette wheel selector* selects individuals from a population based on their fitness.

(8 marks)

- (d) Briefly describe three desirable features of the human immune system that would also be beneficial properties of an artificial immune system.

(6 marks)

- (e) Explain the following terms as they relate to particle swarm optimisation.

i. Particle.

ii. Velocity.

iii. gbest.

(6 marks)

- (f) Perceptrons are simple computational techniques inspired by neurons in the human brain. Explain (with a diagram if necessary) why a perceptron cannot provide correct output for the XOR function and why this led to the stalling of neural network research for more than a decade.

(6 marks)

(Total 40 marks)

Please Turn Over

2. (a) Multi-Objective Evolutionary Algorithms differ in several ways from their single-objective counterparts.

i. Explain what is meant by the term *domination* in multi-objective evolutionary algorithms.

(5 marks)

ii. Describe a process of ranking solutions within a population using the domination principle and explain why this is necessary in multi-objective optimisation.

(5 marks)

iii. Explain why multi-objective optimisation is preferable to repeating a single-objective optimisation run with different objective weightings.

(5 marks)

(b) Each nature-inspired algorithm has a different method of varying the extent to which it *exploits* or *explores*. For each algorithm below, describe the parameters that are modified to achieve different exploitation and exploration capability and explain how varying these alters the behaviour of the algorithm.

i. Genetic Algorithms

(5 marks)

ii. Particle Swarm Optimisation

(5 marks)

iii. Ant Colony Optimisation

(5 marks)

(Total 30 marks)

Please Turn Over

3. Working with a team of robotics engineers from a Space Agency, you are tasked with evolving a robotic controller using *genetic programming* for an autonomous exploration vehicle (e.g. a Mars rover). The controller must allow the robot to navigate a landscape, using its sensors to avoid obstacles and proceeding towards a goal location (G). The sensors indicate proximity to an obstacle as a binary signal (0=no obstacle, 1=obstacle) in each of the compass points, N, S, E, W and the robot can move in the following manner: Move Forward (F), Turn Left (L), Turn Right (R), Move Backward (B). You are also able to make use of an accurate software simulation of the environment in which the vehicle will find itself.

- (a) Describe the representation that you would use to represent controller solutions to this problem in genetic programming taking care to specify both *functions* and *terminals*.

(6 marks)

- (b) Draw an example tree using your representation and explain what robotic behaviour would result from executing this tree.

(6 marks)

- (c) Describe how you would compute the objective (fitness) function for a controller.

(6 marks)

- (d) Describe a set of experiments that would be conducted to determine which parameter settings are optimal for the genetic programming algorithm. In your answer make sure that you describe the parameter being modified, any bounds that may exist for the parameter setting and a method for gauging the performance of the algorithm.

(6 marks)

- (e) The Agency informs you that a new set of sensors are available that can measure the distance between the robot and an obstacle and return this distance as a floating-point value. Explain any modifications you would make to your initial representation described above to work with the new sensor readings.

(6 marks)

(Total 30 marks)

Please Turn Over

4. Neural networks are useful pattern recognition tools and can also be used to study human cognition.

- (a) Describe the neural network approach (including representation), and briefly the experimental findings in the paper Sejnowski, T. J. and Rosenberg, C. R. (1987) "Parallel networks that learn to pronounce English text" *Complex Systems* 1.

(15 marks)

- (b) Self-organising maps (also known as Kohonen networks) are neural networks that are able to learn structure from data using unsupervised learning.

- i. Explain what is meant by the word *unsupervised* in this context and how *unsupervised learning* differs from *supervised learning*.

(5 marks)

- ii. Show, with the aid of equations, how a self-organising map learns from an input pattern presented to it.

(5 marks)

- iii. Two variables involved in the learning process in a self-organising map will change over time. Explain what these variables are, how they change, and the effect this has on the learning process.

(5 marks)

(Total 30 marks)

End of Paper